

Marking rubric:

Q 1	a (i)	a (ii)	c	Q 2	a (i)	a (ii)	c	Q 3	a	b (i)	b (ii)
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All above sub-questions are of 1 weightage mark. For each:

Correct Solution: 1 mark, partially correct: 0 – 0.75 mark

Q1:**[2 + 2 + 1 = 5 Marks]**

- a) Consider an IoT-based healthcare system where patient vital signs are sent to a cloud server for analysis.
- Identify which TCP/IP layer(s) is/are most critical for real-time data transmission and justify why.
 - Network Layer: Ensures efficient routing of patient data.
 - Transport Layer: Provides reliability via TCP or speed via UDP.
 - Application Layer: Protocols like MQTT/HTTP handle data transmission.
 - Briefly explain how encapsulation ensures security in this scenario.
Encapsulation alone does not ensure security in TCP/IP. Security can be ensured when encapsulation is combined with other integrity and encryption mechanisms, while sending sensitive medical records.
- b) A circuit-switched network requires a setup time of 450 ms and operates at 15 Mbps. A packet-switched network has no setup time but has a per-packet processing delay of 12 ms per packet. A file of 5 MB is being sent, with each packet size of 1 MB. Compute the total transmission time for both networks.

circuit switch calculation: correct: 1 mark, partial correct: 0 – 0.75**packet switch calculation:** correct: 1 mark, partial correct: 0 – 0.75

File size = 5MB = 5 * 8 Mb,

Setup Time = 450msec = 0.45sec

Processing delay = 12msec = 0.012sec

Circuit Switched

Transmission delay = File size / Transmission rate = 5 * 8 Mb / 15Mbps = 2.67sec

Total delay = Transmission delay + setup time = 2.67 sec + 0.45 sec = 3.12 sec

Packet Switched

Transmission delay for each file = File size / Transmission rate = 1MB / 15Mbps = 0.53sec

Total delay = (Transmission delay + Processing delay) for 5 pkts = (0.53 + 0.012) * 5 = 2.72sec

- c) A user queries www.example.com, and the response takes a total of 300msec, because root, TLD, and authoritative DNS servers are involved. Compute the time saved if the TLD server's response is cached locally.

200 msec

Q2:**[2 + 2 + 1 = 5 Marks]**

- a) A browser loads a webpage with 10 objects from the same server.
- Compute the total time needed if using non-persistent HTTP with 2 parallel TCP connections and an RTT of 100 ms.
Non-persistent HTTP:
2RTT for each object:
Total time: 10 objects * 2 RTT * 100 ms = 2000msec = 2 sec
For 2 parallel connections: 2/2 = 1 sec.
 - How does persistent HTTP with pipelining improve performance?
Persistent HTTP establishes connection for one time, and then only 1 RTT for each object.

- b) A web server handles 3600 requests per hour, with each requested object averaging 500Kb in size. When a caching proxy is introduced, as a result, the hit ratio improves to 80%. Compute the bandwidth saved per hour due to caching.

Total bandwidth calculation: correct: 0.5 mark, partial correct: 0 – 0.25

Non-Cached bandwidth calculation: correct: 1 mark, partial correct: 0 – 0.75

Total bandwidth saved calculation: correct: 0.5 mark, partial correct: 0 – 0.25

Total bandwidth = $3600 * 500\text{Kb} = 1.8 \text{ Gb}$

With 80% cache hit, server has to handle only 20% requests per hour: $3600 * 0.2 = 0.36 \text{ Gb}$

Total Bandwidth saved = $1.8 \text{ Gb} - 0.36 \text{ Gb} = 1.44 \text{ Gb}$

- c) In addition to HOL blocking mitigation, and priority based transmission, what is the main factor that resulted in HTTP/2 to provide better efficiency than HTTP1.1.

Push unrequested objects

Q3

Time Estimate: 15 minutes

[1 + 2 + 2 = 5 Marks]

- a) What is the function of the twisting in twisted-pair cable?

To cancel electromagnetic interference

- b) Consider the scenario shown in Figure. 1, in which transmission rates and propagation delay at each link are shown. Assume there is no cache, there is no queuing delay at the routers, and the packet processing delays at routers and nodes are all zero. All packets in the network are 10,000 bits long.

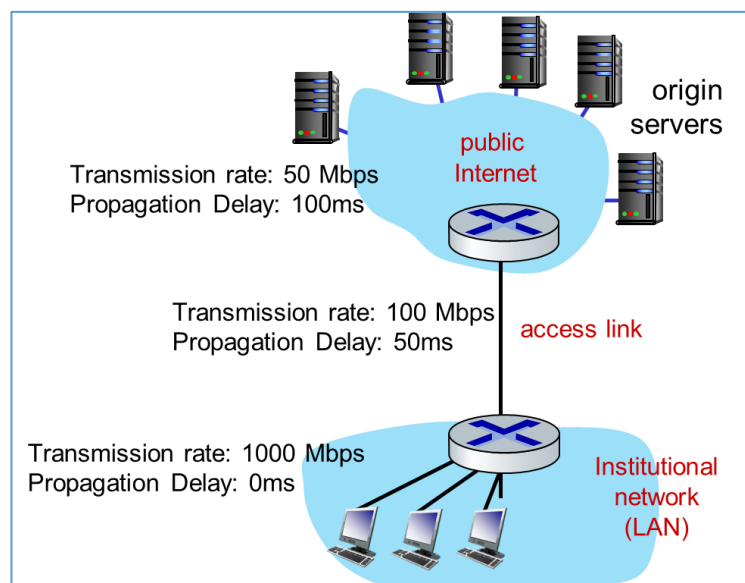


Figure 1. A network scenario

- i. What is the end-to-end delay from when a packet is transmitted by the server to when it is received by the client?

Transmission delays = File Size / Transmission rate

Find transmission delays of all nets, and then sum them up all, including Propagation delays

- ii. What is the maximum rate at which the server can deliver data to a single client if we assume no other clients are making requests?

Bottleneck link = 50Mbps

c) Redraw the below Table in your answer script, and complete it with appropriate entries.

Each entry of table: correct solution: 0.0125 marks, otherwise: 0

Layer Name	PDU	Device	Protocols	Addressing
Application	Message	PC / Laptop	HTTP, DNS	Specific address (URL)
Transport	Segment	Firewall/Gateway	TCP, UDP	Port address
Network	Datagram	Router	IP	Logical (IP) address
Data Link	Frame	Switch	Ethernet / PPP	Physical address

===== Thank you =====